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ORIGINAL ARTICLE

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Human-AI collaboration: Designing artificial agents to facilitate socially shared regulation among learners

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Abstract: Socially shared regulation of learning (SSRL) is a crucial process for groups of learners to successfully collaborate. Detecting and supporting SSRL is a challenge, especially in real time, but hybrid intelligence approaches such as Artificial Intelligence (AI) agents may make this possible. Leveraging the concept of trigger events which invite SSRL, we present a design of an AI agent, MAI, which can detect SSRL and prompt students to raise their group-level metacognitive awareness with the aim of facilitating SSRL. We present the methodology we used to design MAI, drawing on the Echeloned DSR (eDSR) Methodological Framework and making use of the Wizard of Oz prototyping paradigm. We likewise present empirical results evaluating our initial prototype of MAI, using lexical alignment between speakers as a quantitative measure of the effect of MAI's prompts on facilitating SSRL, the Partner Model Questionnaire as a quantitative measure of perceptions of MAI, and interviews as qualitative context for these perceptions. We found that the first prototype of MAI did not facilitate SSRL as hoped, possibly owing to mixed perceptions of MAI's reliability and lack of clarity about MAI's role in the collaborative learning task. From these findings, we offer revised prompts for the next iteration of prototyping this agent and a refined

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set of design requirements for future development of metacognitive AI agents for supporting SSRL.

KEYWORDS
artificial intelligence, collaborative learning, design science research, human-AI collaboration, socially shared regulation, wizard of Oz

Practitioner notes

What is already known about this topic

- Socially Shared Regulation of Learning (SSRL) is recognized as a critical component for the success of collaborative learning, emphasizing the importance of group-level regulatory processes in achieving shared goals, enacting strategies and monitoring learning progress.
- Supporting SSRL in face-to-face collaborative learning environments presents challenges, including the complexity of coordinating and synchronizing individual contributions and regulatory actions within a group context.

What this paper adds

- This paper introduces the design of Metacognitive Artificial Intelligence (MAI), a novel AI system aimed at enhancing Human-AI collaboration for supporting and augmenting SSRL processes.
- Through empirical research, the study offers lessons learned and design considerations for developing artificial agents on facilitating and enhancing SSRL among learners, demonstrating how AI can play a pivotal role in collaborative learning environments.
- The findings highlight the critical importance of multidisciplinary knowledge in the design of multi-agent interfaces (MAI) that provide real-time, adaptive support for group metacognitive processes and decision-making.

Implications for practice and/or policy

- Educational technologists can utilize the proposed design principles in the development and integration of MAI tools to enhance SSRL.
- Educators can incorporate the principles of MAI and our relevant findings into their teaching strategies to actively foster and support socially shared regulation of learning among students.
- Policymakers should consider revising educational frameworks to include the use of AI technologies that support SSRL strategies in collaborative learning.

INTRODUCTION

In the age of AI, learners' agency should drive human-AI collaboration. Self-regulated learning (SRL) refers to an agentic process whereby learners strategically take control of their learning engagement (Zimmerman, 1989). Socially Shared Regulation of Learning (SSRL), in turn, is a critical component for the success of collaborative learning (Järvelä et al., 2018),

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involving a complex process where learners collectively manage their learning activities, influencing each other's cognitive, motivational, and emotional states (Haataja et al., 2022; Kent & Cukurova, 2020; Zheng et al., 2019). The role of educators in fostering SSRL competencies is vital, yet the global scale of learning necessitates technological interventions to enhance SSRL among learners. The integration of Artificial Intelligence (AI) in various aspects of life and work further emphasizes the importance of developing learners' abilities to collaborate not only with peers but also within socio-technological systems involving AI.

Despite the recognized potential of multimodal data and AI-driven techniques in supporting SSRL, efforts to design systems that leverage these technologies are limited. The development of a comprehensive AI-enhanced system to facilitate SSRL also represents a formidable challenge stemming from the complexity of designing systems that are both theoretically sound and technologically advanced, capable of addressing the multifaceted nature of SSRL. Theoretically, the conceptualization and operationalization of SSRL are hindered by the difficulty in observing the emotional and cognitive processes central to regulatory activities. Although advances in multimodal data and techniques offer new insights into SSRL, existing frameworks fall short as they were developed before these methods became prevalent. Recently, Järvelä et al. (2023) introduced a trigger concept and the HASRL model, utilizing multimodal data and advanced technologies to support SSRL, marking a significant theoretical advancement. However, continuous efforts are required to explore SSRL triggers and guide effective design.

Technologically, creating a comprehensive AI-enhanced system to support SSRL is both expensive and time-consuming, posing significant risks for researchers engaged in design science research (DSR). Fortunately, the echeloned design science research (eDSR) methodology, as proposed by Tuunanen et al. (2024), emerges as a pivotal response to these challenges. This methodology distinguishes itself by providing a structured, phased approach to DSR, enabling researchers to navigate the complexities of system development with greater clarity and direction. By segmenting the research process into distinct, manageable stages, eDSR facilitates focused and iterative exploration of design solutions, allowing for incremental advancements and the systematic evaluation of each development phase. This structured approach not only enhances the efficiency of the research process but also significantly reduces the risks associated with the development of sophisticated AI-enhanced systems for SSRL support.

In addition to the eDSR methodology, the Wizard of Oz (WOz) prototyping paradigm offers a complementary strategy for exploring the potential of AI in educational settings. Widely recognized in the field of Human-Computer Interaction (HCI), the WOz paradigm simulates the functionality of AI systems through human intervention, thereby eliminating the immediate need for fully operational technological solutions. This approach enables researchers to conduct exploratory studies on system interactions and user experiences without the extensive resources typically required for developing complete AI systems. By employing the WOz technique, researchers can gather valuable insights into the design and functionality of AI-enhanced learning environments, informing the development process and ensuring that the final system is both effective and aligned with the needs of learners.

Despite the potential benefits offered by the echeloned eDSR methodology and the WOz technique for designing complex educational technologies, there remains a notable gap in the literature regarding their practical application (Hao & Cukurova, 2023; Nguyen et al., 2021). Specifically, detailed examples illustrating how these methodologies can be effectively employed in the development of sophisticated educational technology systems are scarce. This absence of concrete illustrations limits the ability of researchers and practitioners to fully understand and leverage these approaches for the design and evaluation of complex systems aimed at enhancing learning and teaching.

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As a result, this study leverages recent theoretical and methodological advancements to design and evaluate a Metacognitive Artificial Intelligence (MAI) system. The MAI system aims to enhance Human-AI collaboration, supporting and augmenting SSRL processes. By integrating the latest in SSRL theory and DSR methodology, this research contributes to the development of effective AI-driven educational technology systems that foster collaborative learning and regulation among learners. This study seeks to explore two primary research questions: (1) What are the design principles and considerations necessary for developing a Metacognitive Artificial Intelligence (MAI) agent that effectively supports Socially Shared Regulation of Learning (SSRL)? (2) In what ways does the MAI, as an artificial intelligence agent, impact the dynamics of SSRL among learners?

THEORETICAL FRAMEWORK

Socially shared regulation in collaborative learning

Regulatory processes have increasingly been acknowledged in empirical research, extending beyond individual learning to collaborative learning contexts (CL) (De Backer et al., 2022; Emara et al., 2021; Hayashi, 2023; Kielstra et al., 2022; Li et al., 2023; Sobocinski et al., 2022; Zheng et al., 2019). Informed by Winne and Hadwin's (1998) four-phase model of SRL, SSRL unfolds unfolding over four loosely sequenced and recursive phases (see Järvelä et al., 2018). Throughout these regulatory cycles, collective monitoring and evaluation emerge to guide group decision-making and adaptation of collaborative processes, progress, and products, thereby intentionally optimizing learning where needed. Importantly, theory emphasizes purposeful, timely practices and strategies deployed in response to situated disruptions or benign situations in learning. Therefore, Järvelä and Hadwin (2024) introduce a trigger regulation framework for advancing multimodal analytical approaches to research about SRL and SSRL. Events and/or situations that may inhibit learning processes and thus require regulatory responses are defined as trigger events. Theory guided trigger conceptual framework and HASRL model capacity to leverage AI affordances and multimodal data for SSRL research can provide means for researchers to examine such complex SRL processes which have been hard with the traditional methods and resources. For example, theory emphasizes that SRL and SSRL are all cyclical and recursive processes whereby individuals shape regulatory responses in the moment and in future work. Therefore, timing and sequence of observed strategies and actions have become a focus for SRL research (Molenaar et al., 2021). Coupled with trigger event concept, these data sources provide means for researchers to examine not only the frequency, timing and sequences of regulatory traces (Saint et al., 2020) but identify learners or teams that might be struggling and provide timely intervention or prompts to deploy regulatory strategies as needed.

Human-AI shared regulation in learning (HASRL)

The widespread adoption of AI into various facets of daily life, including the educational sector, necessitates a reevaluation and revision of current theoretical perspectives to accommodate the integration of AI and human intelligence in education (Gašević et al., 2023; Nguyen et al., 2024). This integration is pivotal for fostering a synergistic relationship that enhances learning outcomes. In response to this emerging landscape and based on the trigger concept framework, Järvelä et al. (2023) introduce the model of Human-AI Shared Regulation in Learning (HASRL) as a foundational framework to address this need. This innovative framework proposes a structured approach to understanding and developing the

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interactions between humans and AI in learning environments, emphasizing the importance of shared regulatory processes.

HASRL is also designed to explain the mechanisms of shared regulatory processes between humans and AI, offering guidance for the design and development of socio-technological systems that facilitate such collaboration. HASRL identifies crucial elements within both human and AI regulatory processes and their interactions, advocating for systems that are responsive not only to initial triggers but also adaptable in tuning AI models to reinforce human regulatory capabilities (Järvelä et al., 2023). This approach suggests a move toward educational technologies that actively support human learning through AI, thereby enriching the learning experience.

Although HASRL is presented as a progressive framework for the development of socio-technological systems that enable a productive partnership between human and artificial intelligences, Järvelä et al. (2023) highlight the need for empirical and interdisciplinary research to further refine HASRL, which is fundamental for crafting effective systems that encourage shared regulation and enhance collaborative learning with AI. Therefore, this study adopted HASRL as its theoretical framework, employing DSR methodology from information systems research alongside the WOz approach from the field of human-computer interaction, to explore the design of AI systems that support SSRL within a collaborative learning context.

RESEARCH METHODOLOGY

Design Science Research (DSR) is a widely used methodology in information systems, valued for its dual focus on creating information system artefacts and advancing both specific and general design theories (Hevner et al., 2004; Peffers et al., 2007). This methodology aligns with studies that explore human-AI shared regulation and aim to develop AI tools to effectively support SSRL in collaborative learning. However, DSR has faced criticism for being resource-heavy and lengthy, which can affect the feasibility of research projects and negatively influence the accumulation of knowledge and the thorough evaluation of research findings. To address these concerns, recent methodological developments have introduced the Echeloned DSR (eDSR) Methodological Framework (Tuunanen et al., 2024), offering a more dynamic approach to conducting DSR. Moreover, in human-AI interaction, the WOz technique is recognized for its efficiency in early testing of design and interaction without substantial investments (Schlögl et al., 2015). This method enables a detailed exploration of design aspects. As a result, this study employs the eDSR methodology along with the WOz technique to guide an efficient yet comprehensive investigation into designing AI for supporting SSRL in collaborative learning contexts.

Echeloned DSR (eDSR) methodological framework

The eDSR methodology introduces a novel approach to DSR designed for managing the inherent complexities within multifaceted design research projects (Tuunanen et al., 2024). Developed as an advancement over traditional DSR methods, this framework employs a hierarchical, multi-echelon organizing logic that accommodates complex socio-technical systems and varied stakeholder viewpoints. Unlike conventional DSR approaches that typically progress linearly, eDSR systematically decomposes a research project into manageable segments called echelons, each focussing on specific aspects of the research process from problem analysis to evaluation.

Figure 1 illustrates the eDSR methodological framework. The methodology is structured around five distinct echelons, starting with the Problem Analysis Echelon, which defines and

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框架：eDSR 方法框架

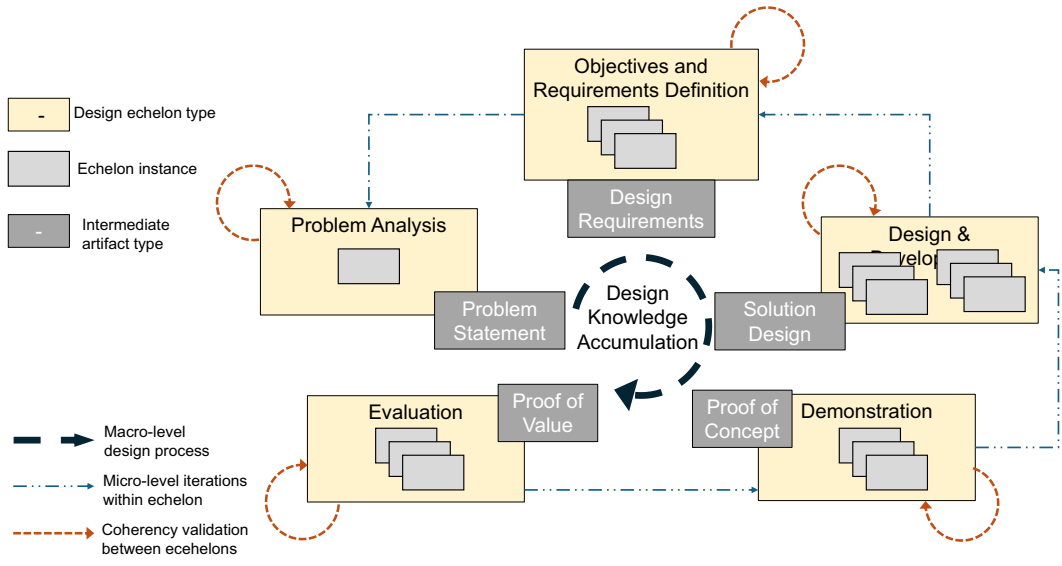


FIGURE 1 Echeloned Design Science Research (eDSR) Methodological Framework (Tuunanen et al., 2024).

refines the research problem. Following this, the Objectives and Requirements Definition Echelon sets specific objectives and requirements for the artefact design. The Design and Development Echelon then takes over, leveraging iterative processes to create the artefact. This is followed by the Demonstration Echelon, which showcases the artefact's functionality in a simulated or real-world environment, and finally, the Evaluation Echelon rigorously assesses the artefact's efficacy and utility in its intended context.

The adoption of an echeloned approach is motivated by the challenges encountered in traditional DSR, such as the difficulty of managing research complexity and maintaining relevance and rigour amidst dynamic problem and solution spaces. The eDSR framework addresses these issues by enhancing manageability through the segmentation of the research process into smaller, logically connected phases that enable focused problem-solving and validation efforts. Each echelon incorporates specific validation points, adopting a “fail early, fail often” philosophy that facilitates early risk mitigation. Moreover, the flexibility and adaptability of the eDSR methodology make it suitable for various research contexts, allowing researchers to tailor the echelon structure to meet the specific needs and dynamics of their projects. This structured yet flexible framework ensures continuous stakeholder engagement, keeping the research aligned with user needs and real-world applicability.

In this study, we concentrate on the Objectives and Requirements echelon, aiming to produce design requirements for HASRL based on the problem statement derived from previous research. Additionally, we contribute other intermediate artefacts, including a solution design and a proof of concept. These elements are developed to ensure that the final outputs align closely with the identified needs and challenges, reinforcing the practical relevance and applicability of the research.

Wizard of Oz (WOz) approach to educational technology design

The WOz paradigm is a prototyping paradigm which has been used to study interactions between research participants and technological artefacts for more than four decades (Schlögl et al., 2015). Largely used to study natural language interactions with computers,

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the technique approach emerged to understand how people would interact with machines which understood speech before this technology was feasible to create (Gould et al., 1983). In a WOz experiment, a participant is told that a piece of technology will have particular capabilities, such as being able to understand their speech and respond to it. In reality, a researcher simulates the machine's abilities, for example by listening to participants' speech and playing recorded audio to create an illusion that the machine is interacting in this way autonomously.

The WOz approach has gained popularity in prototyping of educational technology designs, especially as a response to challenges in developing and understanding emergent technologies such as multimodal AI (Cukurova et al., 2020). For example, a recent study investigating learners' usage of AI-generated summaries of discussions relied on this approach for generating summaries, enabling better understanding of a present or near-future technological use case (Hao & Cukurova, 2023). When using this approach, researchers typically set constrain Wizard behaviour to the expected behaviour of the real, near-future technology that they are simulating (Riek, 2012). Likewise, Wizards are trained on the way they are meant to interact, ensuring that they will follow the intended constraints during interactions with participants (Riek, 2012). Iterative design can therefore occur rapidly, with insights from a demonstration feeding back into design and development and into requirement definition before the completion of technical development. In this way, moving between echelons in an eDSR framework is faster, with more design iterations contributing to each successive demonstration.

Data gathering and analytical methods for demonstrating and evaluating MAI

For demonstrating and evaluating MAI, we adopt the evaluation criteria suggested by Venable et al. (2012) to: (a) Evaluate the artefact to establish its utility and efficacy for achieving its stated purpose; (b) evaluate the artefact to identify weaknesses and areas of improvement; and (c) Evaluate the artefact to identify side effects or undesirable consequences of its use.

Participants and procedure

Fifty-two pre-service teachers participated in this study. They were divided into 14 groups of 3–4 members for 60 minutes collaborative inquiry learning task on the topic of forms and conservation of energy. We created a collaborative learning script comprised of an introductory video to the topic (5 minutes); simulation with a set of guiding questions and answer sheet (University of Colorado Boulder, 2024; 20 minutes); and an experimental task that focused on the energy forms and energy conservation when different balls bounced on the floor (25 minutes). At the end of the session, students had reflective discussion about using these kinds of learning activities in their teaching practices (10 minutes). We implemented all the learning activities and their instructions into an inquiry learning space of the GoLab learning environment (Figure 2; Golabz, 2020).

During the task, the students engaged with MA, in two modalities. They could ask task-related questions via text messages through a chatbot interface that utilized OpenAI's GPT-4 which we prompted to avoid providing direct answers to questions and to instead emphasize reflection and foster finding alternative strategies to solve problems within groups. MAI also listened to the groups' collaboration and suggested SSRL prompts via a speaker to help them proceed in their group work. To control for inaccurate detection of speech in authentic

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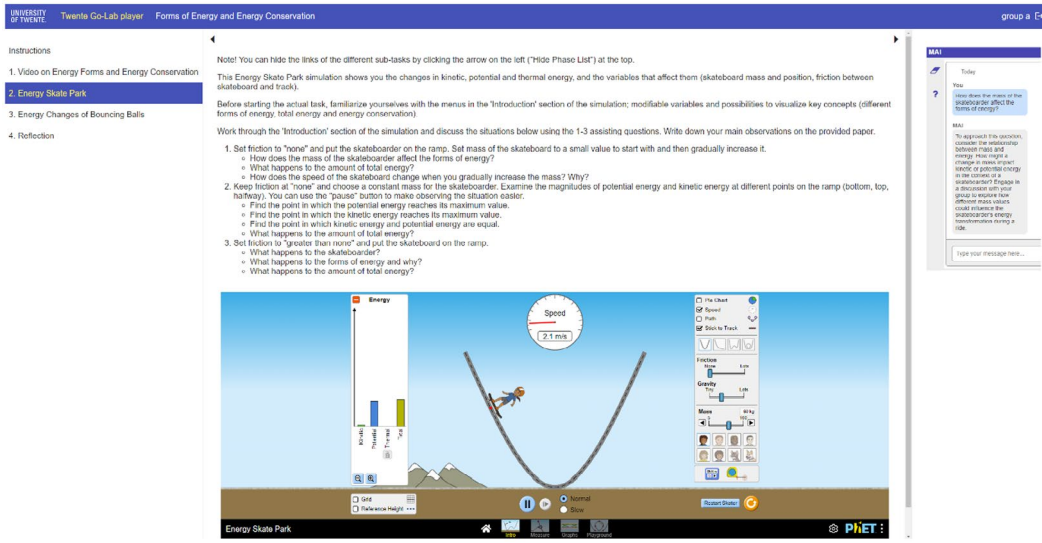


FIGURE 2 Screen-capture of the GoLab learning environment that included instructions and resources relating to the collaborative inquiry learning task: Simulation with a set of guiding questions (in the middle), MAI chatbot interface (on the right) and navigation panel (on the left).

collaborative learning situations, we utilized the WOz paradigm for simulating this interaction modality.

Data analysis

Video and audio were collected to evaluate the SSRL process of learners through interactions with MAI prototype. Group interviews were conducted at the end of the session to capture learners' perception on MAI. The Partner Model Questionnaire (Doyle, 2022) was administered before and after the interaction to evaluate learners' perceptions of MAI.

Linguistic alignment analysis

To understand how MAI affected the way learners communicated and coordinated their language, we measured the lexical alignment among groups before and after interventions from MAI. Alignment is an automatic process by which speakers in conversation with one another reuse linguistic forms, building a shared representation of their situation (Pickering & Garrod, 2004). Prior empirical work has shown alignment across several linguistic levels, including at the lexical level of word choices (Brennan & Clark, 1996) and syntactic level of sentence structures (Branigan et al., 2000).

Recently, collaborative learning research has investigated linguistic alignment to measure the effectiveness of collaboration. Recent collaborative learning research has shown that alignment can be a useful indicator of the development of mutual understanding throughout a collaborative learning task (Hayashi, 2023). Likewise, recent research has explored the measurement of various levels of alignment during collaborative problem solving, finding levels of alignment indicative for differentiating phases of collaborative learning (Buseyne

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et al., 2024). By measuring lexical alignment before and after prompts from MAI, we aimed to investigate whether prompts were effective in raising group-level metacognitive awareness and allowing learners to progress with their collaborative learning tasks.

Partner modelling questionnaire

Speech interaction research introduced the concept of partner models, the mental representations people make of the capabilities of their conversational partners, whether those partners are other people or machines (Branigan et al., 2011; Cowan et al., 2015). A validated measure of this concept for human-computer speech interactions has recently been introduced, assessing people's representations machines' conversational capabilities: the Partner Modelling Questionnaire (PMQ) (Doyle, 2022). Whereas prior work in this area relied on making inferences about people's impressions of a machine dialogue partner from their ratings of usability (Hone & Graham, 2001) or satisfaction (Polkosky & Lewis, 2002), the PMQ explicitly measures the partner model construct, giving insights into how the design of an agent impacts users' understanding of its capabilities. The PMQ builds upon prior partner modelling research which identified a three-factor structure to people's formation of partner models of machines, namely: perceptions of partner competence and dependability, assessment of human-likeness, and perceptions of the conversational flexibility of the system (Doyle et al., 2021), enabling researchers to understand the dimensions through which study participants form their partner models of machines. To understand what students thought of MAI's abilities as a conversational partner, we used the PMQ as a pre-test measure (to assess participants' baseline views of conversational agents) and a post-test measure (to compare their views of MAI against that baseline).

Interviews

To get a holistic understanding of learners' impressions of MAI, we also asked groups engaged in a self-directed semi-structure interview by discussing aloud two questions: “How did your collaboration go?” and “How do you think MAI (an AI agent) has been useful or not useful in your collaboration?” Questions were presented on a sheet of poster paper with the instructions to discuss both questions aloud, and participants were free to discuss for up to 10 minutes. This reflection was intended to give further context to quantitative measures of the impact of MAI, allowing for insights into what design features helped or hindered MAI achieving the intended goals.

DESIGN REQUIREMENTS AND PRINCIPLES FOR METACOGNITIVE ARTIFICIAL INTELLIGENCE (MAI)

Following eDSR framework, we begin with the Problem Statement defined from previous studies, which sets the stage for our initial design requirements. Building on these requirements, we have developed a Wizards of Oz prototype to demonstrate and evaluate the feasibility and functionality of our proposed solution in a real-world setting. This approach allows us to systematically explore and refine our design requirements based on direct feedback and interaction within the intended context.

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Problem statement

SSRL has extensive empirical evidence supporting its critical role in enhancing collaborative learning outcomes (eg, Hadwin et al., 2018; Khosa & Volet, 2014; Molenaar & Järvelä, 2014). Despite its recognized importance, numerous studies have highlighted a significant deficiency in the implementation of SSRL within collaborative learning environments. Often, students overlook the crucial aspects required for SSRL, which could otherwise significantly enhance the efficacy of collaborative learning. While the potential for educational technological solutions to support SSRL is widely acknowledged and emphasized, there is a notable absence of comprehensive guidance and principles for the design and development of such systems. These systems must be well-grounded in learning theories and proven effective in practice, yet current literature and practice provide limited direction on how to achieve these criteria effectively. This gap underscores the need for a structured and evaluated requirements and principles that guides the creation of technological solutions aimed at facilitating SSRL in educational settings.

Initial design requirements and principles

We adopted the set of design principles for supporting SSRL from Järvelä et al. (2015) as the initial design requirements and principles for designing MAI system. These principles serve as foundational guidelines to ensure that the system effectively facilitates and enhances collaborative learning experiences, enabling users to engage actively in regulating both their individual and shared learning processes.

- DP1. Awareness: MAI should increase learners' awareness of their own and others' learning process.
- DP2. Externalization: MAI should support the externalization of students' and others' learning process in a social plane and helping in sharing and interaction.
- DP3. Prompting regulation: MAI should prompt the acquisition and activation of regulatory processes.

First, increasing learners' awareness of their own and others' learning processes is fundamental, as this awareness is crucial for metacognitive engagement, allowing learners to monitor and evaluate their strategies and progress. Second, supporting the externalization of students' and others' learning processes on a social plane enhances collaborative opportunities, enabling the sharing and interaction that are vital for co-constructing knowledge and skills. Lastly, support tools must prompt the acquisition and activation of regulatory processes, equipping learners with the necessary skills to manage their learning actively and adaptively in social contexts.

Wizards of Oz prototype

The WOz prototype was designed to demonstrate interactions between collaborative learners and MAI, providing a proof of concept. This prototype simulates the presence of an autonomous proactive speech agent (ie, a speech agent which select when to interact rather than waiting for an input from a user) embedded within an iPad on the students' table, which appears to participate actively in conversations. The prototyping approach is illustrated in Figure 3.

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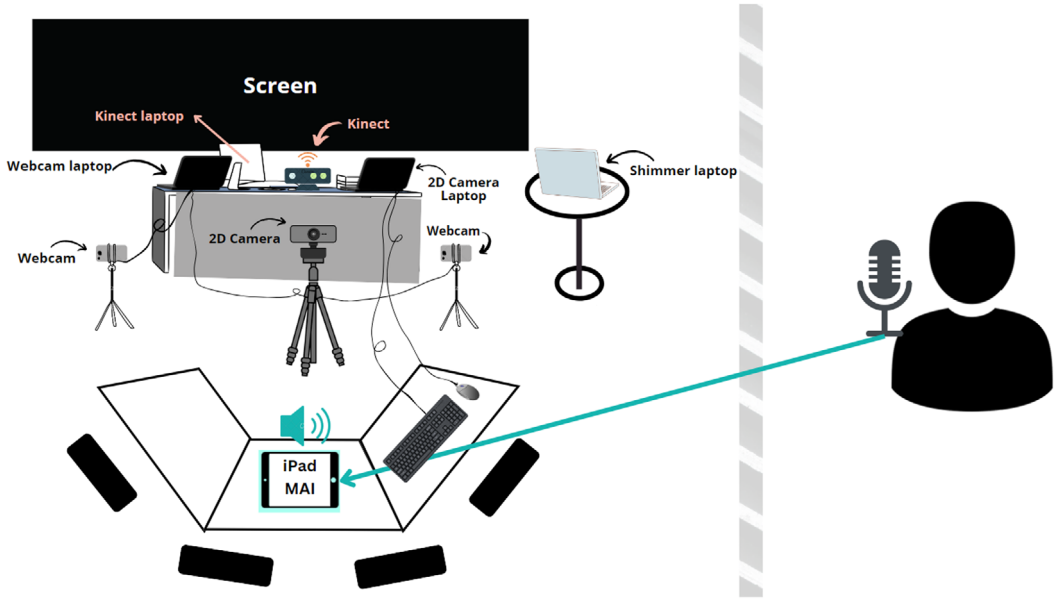


FIGURE 3 Illustration of the Wizard of Oz prototype approach used in the study.

The researchers playing the role of Wizards were in a room adjacent to the learning task, where they could listen to a group's discussion. Wizards were trained to simulate the agent's ability to understand spoken language and apply basic if-then rules for responding, demonstrating a technological artefact which is already feasible to develop, but at a reduced resource cost at this stage of iterative design. Wizards listened to groups' conversations and then played a pre-recorded, randomly selected intervention from the list in Figure 4. Interventions were recorded by a native Finnish speaker. Prompts were selected according to the rules described in Figure 4.

DEMONSTRATION AND EVALUATION OF HUMAN-AI COLLABORATION WITH (MAI)

The design requirements and principles embedded in the prototype of the MAI were evaluated using a cohort of 52 pre-service teachers, as detailed in the above research methodology section. This evaluation aimed to assess the effectiveness and functionality of the prototype in actual collaborative learning settings, ensuring that it met the anticipated educational needs and interaction criteria established for this technology.

Influence of MAI on linguistic alignment

Following prior research on alignment, lexical alignment between group members was measured across two categories of words: the 25 most commonly used words across the discussion, which proxies for general alignment across a conversation (Friedberg et al., 2012), and the words from the task description of the task the group was working on when they were prompted, as task description work alignment has been shown to be related to collaborative problem solving success (Rahimi et al., 2017). Alignment between a pair of speakers for a word (w) in each category was calculated using the following formula, where ALL is the total

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框架：MAI 提示規則

PRIORITY: First check blue → if not → check green → if not → check red
NEGATIVE VALENCE MONITORING COGNITION **IF** group members use keyword "I/we don't../can't" 3 times consecutively (NOTE! NO "I DON'T UNDERSTAND") **THEN** prompt:
I noticed that you are facing a difficulty… • Can you describe what is going on right now? Do you all agree? • What might clarify this concept/issue for you? • How could you help yourself progress? • How is your current approach helping you achieve your goals?
ISSUE RELATED TO CONCEPTUAL UNDERSTANDING IF what group members express is semantically similar to confusion and potentially on same target (i.e., domain knowledge) in one episode, over 3 turns of speaking, **THEN** prompt:
I noticed that you might be confused… • Can you explain what are you trying to understand now? • Do you all understand your task the same way? How about the key concepts? • How confident do you feel that you have the tools to complete your task? What would help you?
LACK OF SHARED PERSPECTIVE: IF group members use first or second person singular ("I"/"You") three times in a row **THEN** prompt:
I noticed that you are thinking of your individual perspectives… • How can you progress as a group? • What is your joint plan for solving the task? • What are you trying to accomplish here as a group?

FIGURE 4 MAI prompting rules.

number of words from the respective speaker and count(w) is the number of times word w is spoken:

$$\text{align}(w) = -\left|\frac{\text{count}_{s1}(w)}{\text{ALL}_{S1}} - \frac{\text{count}_{s2}(w)}{\text{ALL}_{S2}}\right|.$$

The equation is negated, so that pairs of speakers with higher levels of alignment approach 0, representing perfect alignment for a given word, while perfectly misaligned speakers on a given word have scores approaching −1. As in alignment research involving groups of more than two speakers (Friedberg et al., 2012; Rahimi & Litman, 2018), align(w) is computed for each category (c) of words by summing:

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$$\text{align}(c) = \sum_{w \in c} \text{align}(w).$$

This measure of alignment for a category of words was calculated for each pair of students in each group. Pair scores within a word category were then averaged for that group to attain a single group score per category, as in Friedberg et al. (2012).

Changes in alignment across a group were computed before and after MAI prompts. Because the researchers playing the part of MAI were instructed to wait at least 3 minutes between prompts, we used 3 minutes windows before and after prompts to measure effects on alignment, but results were the same when using windows of 2 or 5 minutes. Because groups varied in how many prompts they received from MAI we only report alignment measures on the first two prompts, as all groups received at least 2 prompts.

To ensure that linguistic alignment was different to the amount expected due to chance, we compared the set of 14 transcripts to a set of 14 randomly shuffled transcripts by which each real utterance in sequence was interleaved with an utterance randomly selected from the real transcripts, following practices from prior work on alignment (Sinclair & Schneider, 2021). Overall, mean alignment was significantly higher for real transcripts than for shuffled transcripts (M $\text{align}_{\text{Real}} = -0.154$, M $\text{align(c)}_{\text{shuffled}} = -0.218$, Wilcoxon rank-sum test $U = 131$, $p < 0.001$, $r = 0.625$), indicating that group members reused the same words more than would be expected due to chance. Surprisingly however, alignment on task description words in any time window was no higher than chance. This may indicate that alignment on these words had little to do with MAI and instead was related to time spent on the task.

We further analysed the differences between alignment before and after prompts for each word category. While alignment on the most common words after the first prompt was greater than expected due to chance, it was not significantly different from alignment on those words before the first prompt (M $\text{align}_{\text{Before}} = -0.805$, M $\text{align(c)}_{\text{after}} = -0.713$, Wilcoxon rank-sum test $U = 167$, $p = 0.098$, $r = 0.313$), indicating that the first prompt from MAI did not significantly increase alignment. Likewise, there were no significant differences in alignment on the most common words or on task description words for either the first or second prompt. This result is visualized in Figure 5.

結論：提示前後語言對齊（RQ2）

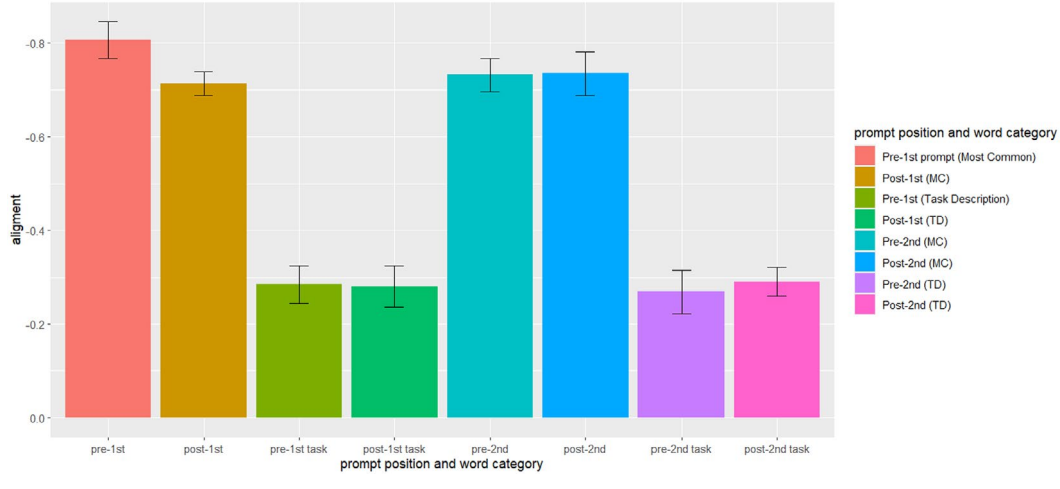


FIGURE 5 Groups' linguistic alignment in the most common words and task description words before and after MAI prompts.

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Learners' perspective on MAI—PMQ and interview excerpts

PMQ scores were not normally distributed, so pre and post-test scores were compared using non-parametric tests. We found a significant difference between pre-test measures of partner models of speech agents as compared to measures of MAI as a conversational partner (Wilcoxon signed-rank test $V = 1017$, $p < 0.01$), with participants' median PMQ score for the pre-test (76) higher than the median post-test PMQ score (64), indicating that they believed MAI to be a weaker conversational partner than their prior expectations for speech agents. This difference is visualized in Figure 6.

To better understand the cause of the difference between pre and post-test scores, we also compared scores on each of the PMQ's three subscales: competence and dependability, human-likeness, and conversational flexibility. While we found no significant differences between pre and post-test scores on human-likeness or conversational flexibility ratings, there was a significant difference between pre-test measures of the competence and dependability of speech agents as compared to measures of MAI's competence and dependability (Wilcoxon signed-rank test $V = 1086$, $p < 0.001$), with participants' median PMQ score for the pre-test (47) higher than the median post-test subscale score (36). This difference is visualized in Figure 7.

To contextualize learners' impressions of MAI, we also examine some illustrative excerpts from the reflective interviews. Group 2 highlighted MAI's negative presence, as they saw prompts as highlighting critiques without helping the group to make progress (Figure 8):

Interestingly, one group member suggests that MAI might have been helpful if the group needed support in getting a group member to speak more or in making group contributions more equal. Insofar as recognizing a lack of shared perspective was one of MAI's aims, the

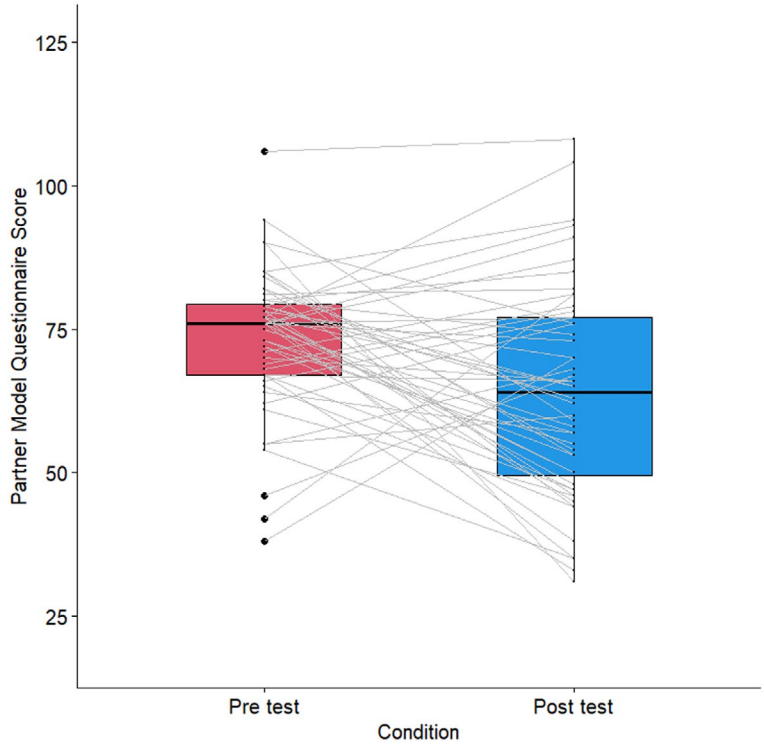


FIGURE 6 Pre and post-test PMQ scores (range 18–126). Lines connect individual participants' paired scores.

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次要：PMQ 前後測

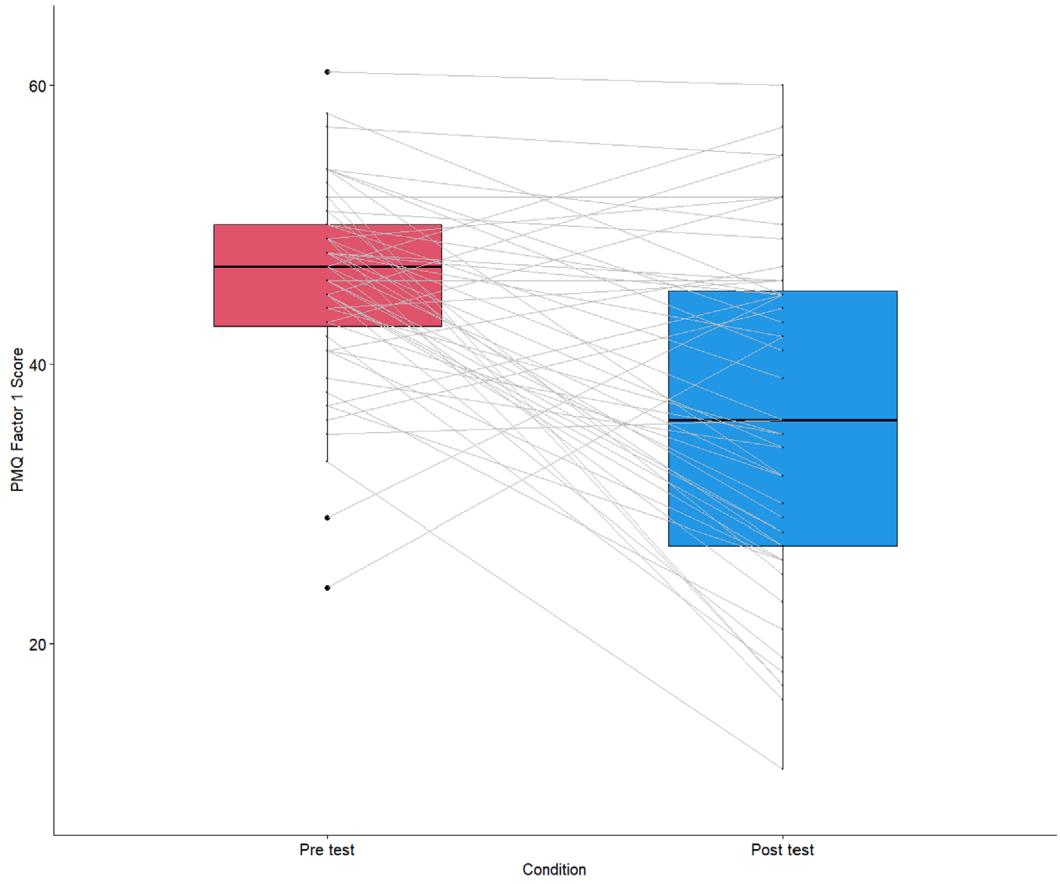


FIGURE 7 Pre and post-test PMQ scores (range 9–63). Lines connect individual participants' paired scores.

group did identify this as a potential opportunity for MAI, but not one that they found beneficial to their collaboration (Figure 9).

Others saw an initial benefit to MAI, mentioning that the first prompt they received was helpful, particularly in raising awareness of the perspectives of others. This provides evidence that DP1 and DP3 were successfully achieved. Still, even this group found that additional prompts weren't very helpful. Likewise, this group had mixed feelings about MAI understanding their conversation and then prompting based on this context, with one group member calling this attribute nice and another calling it weird (Figure 10).

Still, other groups had more positive impressions of MAI. Group 11 identified that MAI helped to resolve differences in understanding between group members, making their collaboration smoother. Likewise, this group rightly identified that MAI did not give direct information about the content of the learning tasks, instead directing learners to reflect on their own thinking, which they see as a strength rather than a weakness of MAI's design.

LESSONS LEARNED AND REDIFINED SET OF DESIGN REQUIREMENTS AND PRINCIPLES FOR MAI

Quantitative and qualitative data analysis of learners who interacted with our prototype version of MAI reveals that MAI was not entirely successful in achieving our intended objectives.

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G2

P: Discuss about MAI
Y: We did throw questions at it, but no.
G: MAI was not much help.
B: MAI really caused me nothing but frustration
Y: It was of no use at all.
B: It was just shouting things. I mean, it felt like someone in a group...
...we have been discussing, like someone in a group work, who just criticizes and does nothing to actually help...
...so it was like, it gave the same amount of input.
G: It was just confusing.
P: But I don't know, if MAI would be useful for...
...for example, if there was like a group, who wouldn't dare to speak...
...or something. Or if we were just quiet the whole time...
...or if there's just one person who speaks, then if...
...it [MAI] can just make a comment as an outsider.

FIGURE 8 An example of a group discussion for which the group had mixed feelings about MAI.

G7

Y: I thought it was quite nice, well no, or funny, that it actually kind of listened to us and we...
...like, it actually realized, that we were confused for example.
B: Yeah.
Y: It must have been taught some certain words, I don't know. But like, the fact that it knew that kind of stuff...
...as if it could spot the problems that it thought we had, which we ourselves did not admit. But.
B: Yeah.
P: It was weird how it sort of took the feelings we had and put it into words.
B: Or it was like, you are confused. Yeah, true.
G: With a monotone voice.
B: Yeah.
P: But then, because we already have a pretty good idea of how we feel, so then we were like oh yeah, weird.
Y: Yup.
P: So.. Like.. I think.. At least I was sort of like, that it didn't really help with my work
Y: It didn't bring anything new to us here.
G: Yeah.
B: But maybe it did, the first time when it said it, that you should discuss or like try to understand each other's points of view.
P: Maybe first time it was good...
G: Yeah, maybe it was a good move…
...but the next two weren't anymore really.
B: Yeah.
G: It mostly sucked.

FIGURE 9 An example of a group discussion for which the group found MAI to be helpful.

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NEGATIVE VALENCE MONITORING COGNITION:

- Hmm, are you feeling a bit stuck? Describe to each other how you understand the problem right now. Do you all agree on how to proceed?
- Good job discussing the task! What could you still do together to get a better understanding of this concept/issue? What might clarify the goals of the task for all of you?
- Good reflecting! How could you still help yourselves progress as a group?
- Try to still jointly discuss how your current approach is helping you achieve your goals. This will help get past tricky situations like this.

ISSUE RELATED TO CONCEPTUAL UNDERSTANDING:

- Am I perhaps sensing some confusion? Explain to each other what you are trying to understand right now.
- While listening to you, I wondered if you all understand the task the same way. Take a moment to discuss what the key concepts are. You can do it!
- How confident are you that you have the tools to complete this task? Try to share thoughts on what might help you move on.

LACK OF SHARED PERSPECTIVE:

- Well done on developing your thoughts! Might you still be thinking about this from your own perspectives? Discuss how you could progress as a group.
- I noticed that you might be thinking about this from individual points of view. I encourage you to discuss what your joint plan is for solving the task.
- Could I help? Rethink what you are trying to accomplish here as a group. By doing this, you can support each other and move on.

FIGURE 10 An example of a group discussion for which the group did not find MAI helpful.

While some saw MAI as beneficial in raising metacognitive awareness at the group level, there was no clear impact on the language behaviour of groups to point toward MAI inviting SSRL. Likewise, participants saw MAI as a relatively weak conversational partner, owing to low competence and dependability. We suggest that these findings are indicative of a need to iterate on the design of both the interaction with MAI and of the overall learning task in which MAI is included, representing a coherency validation feeding back from our proof of concept into the design and development echelon from an eDSR perspective (Tuunanen et al., 2024).

Overall, impressions of MAI were mixed both between groups and between individual learners. While it was surprising that PMQ scores for MAI, a proactive agent which exceeds the capabilities of currently available speech-based conversational agents like Alexa and Siri, was viewed as a weaker communicative partner and less strong in the competence and dependability factor than prior conceptions of speech agents, this finding was in-line with ongoing research of people's impressions of proactive speech agents (Edwards et al., 2024), which likewise found a proactive speech agent to be seen as a weaker dialogue partner and specifically weaker on the competence and dependability factor as compared to either prior impressions of conversational agents. This finding can be seen as a reflection of the novelty of proactivity as a design feature in speech agents, as the novelty of proactive speech may have both positive and negative outcomes. Indeed, PMQ items which load onto the competence and dependability factor such as 'Dependable/Unreliable', 'Consistent/Inconsistent', and 'Reliable/Uncertain' (Doyle, 2022) make clear that novel interactional features like proactivity which are not well-understood by someone who interacts with a speech agent may be seen as unexpected and therefore less strong in this factor.

It may also be the case that proactivity was not what made students see MAI as less competent and dependable, but instead that the particular intervention rules or prompts were the source of this difference. When taken together with linguistic alignment data, which indicates a weak increase in alignment following MAI's first interventions, but diminished effects on later interventions, students seemed to have initially thought highly enough of MAI to change their behaviour following interventions, but then learned to ignore them.

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Likewise, insofar as both alignment and interview data suggest that further prompting beyond the first time MAI speaks may have had diminished effects on collaboration, we suggest that the design of MAI's prompts may have misled learners into believing that MAI repeats the same prompts, even though no prompt was repeated for any group. Indeed, following from prior research, improper setting of mental models for speech agents can lead to a "gulf of expectations" by which interactions fall short of what users hope they will achieve, quickly leading to abandonment of the technology altogether (Luger & Sellen, 2016). Therefore, we propose an additional fourth design principle from the perspective of human-computer interaction, as outlined below:

- DP4. Perception: Interactions with MAI should be designed so that learners perceive it as a competent and dependable.

We also propose revised prompting rules which we plan to test in further iterations of the proof of concept echelon, varying the language used in the beginning of each MAI prompt, indicating to learners that MAI is saying something new every time. These new prompts are presented in Figure 11:

We expect that making MAI's language more varied may reduce the early abandonment that users exhibited in our first demonstration of the agent. Likewise, by making MAI's language more positive and encouraging, we hope to increase learners' perception of MAI's aims—rather than seeing MAI as pointing out flaws, we propose that positively phrased

結論：修訂後提示（RQ1）

G11

Y: I think it went well. Were there any challenges?
P: Maybe when we didn't know about issue.
Y: We didn't understand. Then it was a little unclear sometimes, like isn't that how it is?
P: Yup, that one of us thought differently than the other.
Y: But then luckily MAI said, that you disagree a bit.
B: Helped us.
Y: Your group work. How did it affect your work.
P: It helped to give direction.
Y: Maybe we didn't get stuck so much, and then realized, that now we have to move on.
B: It interrupted to say "do you notice, that you are disagreeing".
P: But then again, it doesn't tell us the right answers...
...it just sort of, told us to think about this. So then we will figure it out ourselves, with the help of it.
Y: Yeah, it helped to support...
...I was maybe thinking something more like...
...that it would have been, a little bit more, giving answers...
...so then it was a bit like, well you're useless, as it gives you more tasks to think about...
...but that kind of broadens the thoughts.
G: It did somehow make the work smoother.

FIGURE 11 Proposed revisions to MAI prompts.

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TABLE 1 Final design principles for Metacognitive Artificial Intelligence (MAI).

#	Design principle	Design principle specification
DP1	Awareness	MAI should increase learners' awareness of their own and others' learning process
DP2	Externalization	MAI should support the externalization of students' and others' learning process in a social plane and helping in sharing and interaction
DP3	Prompting regulation	MAI should prompt the acquisition and activation of regulatory processes
DP4	Perception	Interactions with MAI should be designed so that learners perceive it as competent and dependable

prompts will lead learners to see MAI as both an active listener and as sharing learners' goals of having smooth, productive collaboration (Table 1).

DISCUSSION AND CONCLUSION

This interdisciplinary study investigates the design of a metacognitive agent, specifically MAI, to enhance both the theoretical and practical dimensions of SSRL. MAI is engineered to facilitate SSRL processes in face-to-face collaborative learning environments. Adopting the eDSR methodology (Tuunanen et al., 2024), our research focuses on examining and refining design principles for MAI through the use of a WOz prototype. The study's interdisciplinary approach contributes significantly not only to SSRL theory but also to the methodological advancement within the broader field of educational technology design. By bridging theoretical insights with practical design strategies, this work aims to foster the development of more effective educational technologies that support complex cognitive and metacognitive activities in collaborative learning contexts.

Although the theory-grounded design principles proposed in the domain appear somewhat effective (Järvelä et al., 2015), our results indicate that these are insufficient in real-world contexts, and insights from human-computer interaction are necessary for creating truly effective designs. Learners identifying dissatisfaction about MAI's purpose and abilities, both in interview and PMQ data, may point toward a broader issue around the novelty of proactive agents. The research which developed and validated the PMQ for example suggests that iterative testing of new features like proactive speech is necessary to understand their effects on people's mental models of agents (Doyle, 2022). As such, further comparison of different realizations of proactivity for MAI may reveal insights into how proactive speech can be introduced without causing an agent to seem unreliable. Likewise, recent research on proactive speech agents suggests users should have expectations calibrated through education about the system (Edwards et al., 2024), so further development of MAI may include more robust explanation and demonstration of MAI's capabilities and longitudinal research on users' impressions of MAI across prolonged exposure, through which novel design features become better understood.

There are limitations to the generalisability of the present study to understanding of AI agents in SSRL. First, because the sample is comprised only of higher education students in an education faculty at a single university, wider conclusions about other demographic groups cannot be drawn. Similarly, this study only looked at one AI agent implementation, narrowly focusing on rule-based detection of SSRL trigger events via spoken language. Considering a broader set of modal cues and employing different methods of detection of trigger events such as machine learning approaches would likely lead to different behaviour

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from the agent and interactions from learners. Insofar as the agent learners interacted with had novel proactive speech features, greater familiarity with AI features like this and further development of instructions for learners about agent capabilities will likely result in evolution to this type of learner-AI interaction. Finally, the AI prototype simulated here used the WOz paradigm to expediate testing and control for technical errors in speech recognition, so near-term prototypes of similar agents may face frictions in similar interactions which were not simulated in this study. We therefore encourage future researchers to investigate a wide range of hybrid intelligent interactions through an echeloned design approach, so that we may collectively accumulate richly varied design knowledge in this area.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ETHICS STATEMENT

This research was approved by the Human Sciences Ethics Committee of the University of Oulu (ID 9/2023: Khánh Nguyen) and complied with the university research integrity and ethics guidelines.

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REFERENCES

Branigan, H. P., Pickering, M. J., & Cleland, A. A. (2000). Syntactic co-ordination in dialogue. *Cognition*, *75*(2), B13–B25. https://doi.org/10.1016/S0010-0277(99)00081-5

Branigan, H. P., Pickering, M. J., Pearson, J., McLean, J. F., & Brown, A. (2011). The role of beliefs in lexical alignment: Evidence from dialogs with humans and computers. *Cognition*, *121*(1), 41–57.

Brennan, S. E., & Clark, H. H. (1996). Conceptual pacts and lexical choice in conversation. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *22*(6), 1482–1493. https://doi.org/10.1037/0278-7393.22.6.148

Buseyne, S., Rajagopal, K., Danquigny, T., Depaepe, F., Heutte, J., & Raes, A. (2024). Assessing verbal interaction of adult learners in computer-supported collaborative problem solving. *British Journal of Educational Technology*, *55*, 1465–1485. https://doi.org/10.1111/bjet.13391

Cowan, B. R., Branigan, H. P., Obregón, M., Bugis, E., & Beale, R. (2015). Voice anthropomorphism, interlocutor modelling and alignment effects on syntactic choices in human-computer dialogue. *International Journal of Human-Computer Studies*, *83*, 27–42. https://doi.org/10.1016/j.ijhcs.2015.05.008

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Cukurova, M., Giannakos, M., & Martinez-Maldonado, R. (2020). The promise and challenges of multimodal learning analytics. *British Journal of Educational Technology*, *51*(5), 1441–1449.

De Backer, L., Van Keer, H., & Valcke, M. (2022). The functions of shared metacognitive regulation and their differential relation with collaborative learners' understanding of the learning content. *Learning and Instruction*, *77*, 101527.

Doyle, P. R. (2022). The dimensions and adaptation of partner models in human-machine dialogue [PhD thesis]. University College Dublin, School of Information and Communication Studies.

Doyle, P. R., Clark, L., & Cowan, B. R. (2021). What do we see in them? Identifying dimensions of partner models for speech interfaces using a psycholexical approach. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. >Yokohama Japan. (pp. 1–14). https://doi.org/10.1145/3411764.3445206

Edwards, J., Doyle, P., Brannigan, H. P., & Cowan, B. R. (2024). Comparing perceptions of static and adaptive proactive speech agents. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces, is Luxembourg City, Luxembourg* (pp. 1–12).

Emara, M., Hutchins, N., Grover, S., Snyder, C., & Biswas, G. (2021). Examining student regulation of collaborative, computational, problem-solving processes in open-ended learning environments. *Journal of Learning Analytics*, *8*(1), 49–74. https://doi.org/10.18608/jla.2021.7230

Friedberg, H., Litman, D., & Paletz, S. B. F. (2012). Lexical entrainment and success in student engineering groups. In *2012 IEEE Spoken Language Technology Workshop (SLT)*. Miami, FL, USA. (pp. 404–409). https://doi.org/10.1109/SLT.2012.6424258

Gašević, D., Siemens, G., & Sadiq, S. (2023). Empowering learners for the age of artificial intelligence. *Computers and Education: Artificial Intelligence*, *4*, 100130.

Golabz: Go-lab initiative | go-lab (2020). https://premium.golabz.eu/about/go-lab-initiative

Gould, J. D., Conti, J., & Hovanyecz, T. (1983). Composing letters with a simulated listening typewriter. *Communications of the ACM*, *26*(4), 295–308.

Haataja, E., Dindar, M., Malmberg, J., & Järvelä, S. (2022). Individuals in a group: Metacognitive and regulatory predictors of learning achievement in collaborative learning. *Learning and Individual Differences*, *96*, 102146.

Hadwin, A., Järvelä, S., & Miller, M. (2018). Self-regulation, co-regulation, and shared regulation in collaborative learning environments. In D. H. Schunk & J. A. Greene (Eds.), *Handbook of self-regulation of learning and performance* (pp. 83–106). Routledge. https://doi.org/10.4324/9781315697048-6

Hao, X., & Cukurova, M. (2023). Exploring the effects of “AI-generated” discussion summaries on learners' engagement in online discussions. In *International conference on artificial intelligence in education* (pp. 155–161). Cham.

Hayashi, Y. (2023). Modeling synchronization for detecting collaborative learning process using a pedagogical conversational agent: Investigation using recurrent indicators of gaze, language, and facial expression. *International Journal of Artificial Intelligence in Education*. *34*, 1206–1247. https://doi.org/10.1007/s40593-023-00381-y

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, *28*(1), 75–105.

Hone, K. S., & Graham, R. (2001). Subjective assessment of speech-system interface usability. In *Proceedings of the 7th European Conference on Speech Communication and Technology (Eurospeech 2001)* (pp. 2083–2086). https://doi.org/10.21437/Eurospeech.2001-491

Järvelä, S., & Hadwin, A. (2024). Triggers for self-regulated learning: A conceptual framework for advancing multimodal research about SRL. *Learning and Individual Differences*, *115*, 102526.

Järvelä, S., Hadwin, A. F., Malmberg, J., & Miller, M. (2018). Contemporary perspectives of regulated learning in collaboration. In F. Fischer, C. E. Hmelo-Silver, P. Reimann, & S. R. Goldman (Eds.), *Handbook of the learning sciences* (pp. 127–136). Routledge.

Järvelä, S., Kirschner, P. A., Panadero, E., Malmberg, J., Phielix, C., Jaspers, J., Koivuniemi, M., & Järvenoja, H. (2015). Enhancing socially shared regulation in collaborative learning groups: Designing for CSCL regulation tools. *Educational Technology Research and Development*, *63*(1), 125–142. https://doi.org/10.1007/s11423-014-9358-1

Järvelä, S., Nguyen, A., & Hadwin, A. (2023). Human and artificial intelligence collaboration for socially shared regulation in learning. *British Journal of Educational Technology*, *54*, 1057–1076. https://doi.org/10.1111/bjet.13325

Kent, C., & Cukurova, M. (2020). Investigating collaboration as a process with theory-driven learning analytics. *Journal of Learning Analytics*, *7*(1), 59–71.

Khosa, D. K., & Volet, S. E. (2014). Productive group engagement in cognitive activity and metacognitive regulation during collaborative learning: Can it explain differences in students' conceptual understanding? *Metacognition and Learning*, *9*(3), 287–307. https://doi.org/10.1007/s11409-014-9117-z

Kielstra, J., Molenaar, I., van Steensel, R., & Verhoeven, L. (2022). Supporting socially shared regulation during collaborative task-oriented reading. *International Journal of Computer-Supported Collaborative Learning*, *17*(1), 65–105.

14678535, 2025, 2, Downloaded from https://bera-journals.onlinelibrary.wiley.com/doi/10.1111/bjet.13534 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Li, X., Hu, W., Li, Y., & Mao, Z. (2023). Exploring what synchronized physiological arousal can reveal about the social regulatory process in a collaborative argumentation activity. *Frontiers in Psychology*, *13*. 1042970 https://doi.org/10.3389/fpsyg.2022.1042970

Luger, E., & Sellen, A. (2016). "Like having a really bad PA": The gulf between user expectation and experience of conversational agents. In *c 427-458*. San Jose, CA, USA. (pp. 5286–5297).

Molenaar, I., Horvers, A., & Baker, R. S. (2021). What can moment-by-moment learning curves tell about students' self-regulated learning? *Learning and Instruction*, *72*, 101206.

Molenaar, I., & Järvelä, S. (2014). Sequential and temporal characteristics of self and socially regulated learning. *Metacognition and Learning*, *9*(2), 75–85. https://doi.org/10.1007/s11409-014-9114-2

Nguyen, A., Hong, Y., Dang, B., & Huang, X. (2024). Human-AI collaboration patterns in AI-assisted academic writing. *Studies in Higher Education*, *49*(5)847-864

Nguyen, A., Tuunanen, T., Gardner, L., & Sheridan, D. (2021). Design principles for learning analytics information systems in higher education. *European Journal of Information Systems*, *30*(5), 541–568.

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, *24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Pickering, M. J., & Garrod, S. (2004). Toward a mechanistic psychology of dialogue. *Behavioral and Brain Sciences*, *27*(2), 169–190. https://doi.org/10.1017/S0140525X04000056

Polkosky, M. D., & Lewis, J. R. (2002). Effect of auditory waiting cues on time estimation in speech recognition telephony applications. *International Journal of Human-Computer Interaction*, *14*(3–4), 423–446.

Rahimi, Z., Kumar, A., Litman, D., Paletz, S., & Yu, M. (2017). Entrainment in multi-party spoken dialogues at multiple linguistic levels. *Interspeech*, *2017*, 1696–1700. https://doi.org/10.21437/Interspeech.2017-1568

Riek, L. D. (2012). Wizard of Oz studies in HRI: A systematic review and new reporting guidelines. *Journal of Human-Robot Interaction*, *1*(1), 119–136.

Saint, J., Whitelock-Wainwright, A., Gašević, D., & Pardo, A. (2020). Trace-SRL: A framework for analysis of microlevel processes of self-regulated learning from trace data. *IEEE Transactions on Learning Technologies*, *13*(4), 861–877.

Schlögl, S., Doherty, G., & Luz, S. (2015). Wizard of Oz experimentation for language technology applications: Challenges and tools. *Interacting with Computers*, *27*(6), 592–615.

Sinclair, A. J., & Schneider, B. (2021). *Linguistic and gestural coordination: Do learners converge in collaborative dialogue?* International Educational Data Mining Society. https://eric.ed.gov/?id=ED615547

Sobocinski, M., Malmberg, J., & Järvelä, S. (2022). Exploring adaptation in socially-shared regulation of learning using video and heart rate data. *Technology, Knowledge and Learning*, *27*(2), 385–404.

Tuunanen, T., Winter, R., & vom Brocke, J. (2024). Dealing with Complexity in Design Science Research: A Methodology Using Design Echelons. *MIS Quarterly*, *48*(2), 427–458.

University of Colorado Boulder. (2024). *PhET interactive simulations: Energy Skate Park*. https://phet.colorado.edu/en/simulations/energy-skate-park

Venable, J., Pries-Heje, J., & Baskerville, R. (2012). A comprehensive framework for evaluation in design science research. In K. Peffers, M. Rothenberger, & B. Kuechler (Eds.), *Design science research in information systems. Advances in theory and practice* (pp. 423–438). Springer. https://doi.org/10.1007/978-3-642-29863-9_31

Winne, P. H., & Hadwin, A. F. (1998). Studying as self-regulated engagement in learning. In D. Hacker, J. Dunlosky, & A. Graesser (Eds.), *Metacognition in educational theory and practice* (pp. 277–304). Erlbaum.

Zheng, L., Li, X., Zhang, X., & Sun, W. (2019). The effects of group metacognitive scaffolding on group metacognitive behaviors, group performance, and cognitive load in computer-supported collaborative learning. *The Internet and Higher Education*, *42*, 13–24.

Zimmerman, B. J. (1989). A social cognitive view of self-regulated academic learning. *Journal of Educational Psychology*, *81*(3), 329–339.

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